# Agenda

1. Announcements
   - HW 7.8 and 9.1 due today
   - Discussion Quiz 4 on Thursday over 7.8 and 9.1
   - We added an extra discussion quiz drop
   - I won't have office hours today;
     I'll have them Monday Feb 27, 1:00-2:30 instead
   - Midterm 1 is next week! go.wisc.edu/1xxzfx

Midterm 1: Tuesday February 28, 7:30-9:00 pm

| Discussion Section | TA Name(s) | Exam Room Assignment |
|---|---|---|
| 321, 323 | Anuk | Van Vleck B239 |
| 322, 324, 325, 326, 327 | Logan, Jiaming, and Jiaqi | Van Vleck B102 |
| 328, 329, 330, 332 | Hongjing and Kevin | Van Vleck B130 |

2. Section 9.3

# Section 9.3 - Separable Equations

A separable equation is a first-order differential equation that can be expressed as $\frac{dy}{dx} = g(x) f(y)$ where $g(x)$ is a function of $x$ and $f(y)$ is a function of $y$.

Activity: Which of the following are separable?

$\frac{dy}{dx} = g(x) f(y)$

1. $\frac{dy}{dx} = xy^2$

$g(x) = x$

$f(y) = y^2$

2. $\frac{dy}{dt} = \frac{8t^3 + 2t}{y^2} = (8t^3 + 2t)\left(\frac{1}{y^2}\right)$

$g(t) = 8t^3 + 2t$

$f(y) = 1/y^2$

3. $\frac{dy}{dx} = e^{7x} + 5y$ inseperable

no product of x and y available

4. $\frac{dy}{dt} = \frac{t}{t^2 y + y} = \frac{t}{(t^2+1)y} = \left(\frac{t}{t^2+1}\right)\left(\frac{1}{y}\right)$

$g(t) = \left(\frac{t}{t^2+1}\right)$

$f(y) = 1/y$

A first-order equation is autonomous if it can be expressed as $\frac{dy}{dx} = f(y)$ where $f(y)$ is a function of $y$.

implied $g(x)$ is 1

(like dx)

# Solving Separable Equations

To solve a seperable diffeq (goal: find y)

$\frac{dy}{dx} = g(x)f(y)$, we:

1. separate the variables → $\frac{dy}{f(y)} = g(x)\,dx$
2. integrate both sides

$$\int \frac{dy}{f(y)} = \int g(x)\,dx$$

3. solve for y

this can be difficult
some HW problems require it ☹
quizzes and exams never will.

**Example:** Solve $\frac{dy}{dx} = xy^2 + 2y^2$

$$\frac{dy}{dx} = y^2(x+2)$$

$$\frac{dy}{y^2} = (x+2)\,dx$$

$$\int \frac{1}{y^2}\,dy = \int (x+2)\,dx$$

$$-\frac{1}{y} = \frac{1}{2}x^2 + 2x + C$$

(optional)

$$\boxed{y = \frac{-1}{\frac{1}{2}x^2 + 2x + C}}$$

**Activity:** Solve $\frac{dy}{dt} = t\sec(y)$

$$\frac{dy}{dt} = t\,(\sec y)$$

$$\int \frac{dy}{\sec y} = \int t\,dt$$

$$\int \cos\,dy = \int t\,dt$$

$$= \sin y = \frac{t^2}{2} + C$$

$$\boxed{y = \arcsin\left(\frac{t^2}{2} + C\right)}$$

## Initial Value Problems

A differential equation is called an initial-value problem if we are given an initial value

**Example:** Solve $\overbrace{\frac{dy}{dt} = \frac{6t}{y}}^{\text{diffeq}}$ where $\overbrace{y(0) = 9}^{\text{data point}}$

$$\frac{dy}{dt} = 6t\left(\frac{1}{y}\right)$$

$$\int \frac{dy}{y} = \int 6t\,dt$$

$$\frac{1}{2}y^2 = 3t^2 + C$$

$$\frac{1}{2}(9)^2 = 3(0)^2 + C$$

$$C = \frac{81}{2}$$

$$\boxed{\frac{1}{2}y^2 = 3t^2 + \frac{81}{2}}$$

$$y^2 = 6t^2 + 81$$

$$y = \sqrt{6t^2 + 81}$$

**Activity:** Solve $\frac{dy}{dx} = e^{-y}(2x-4)$ Where $y(5) = 0$

$$\int e^y\,dy = \int (2x-4)\,dx$$

$$e^y = x^2 - 4x + C$$

$$\cancel{e^{(0)}} = 25 - 4(5) + C$$

$$1 = 5 + C$$

$$C = -4$$

$$y = \ln(x^2 - 4x - 4)$$

$$\boxed{e^y = x^2 - 4x - 4}$$

# Mixing Problems

Mixing Problems refer to a collection of problems where two or more solutions are mixed at various rates

(no salt, just water)
$S(0) = 0$

**Example:** Suppose that a vat contains 100L of water initially. Brine (salt water) enters the vat at a concentration of 5g of salt per liter with a rate of 2L per minute through pipe A. Brine also enters the vat at a concentration of 10g of salt per liter with a rate of 1L per minute through pipe B. The vat solution is well-mixed. Brine leaves the vat at 3L per minute. Find an equation $S(t)$ (grams) for the amount of salt in the vat at time $t$ (minutes).

$10\,\frac{g\text{ salt}}{1\text{ min}}$

$10\,\frac{g\text{ salt}}{1\text{ min}}$

means conc. of salt is uniform.

$\frac{S(t)}{100} \cdot \frac{3L \cdot S(t)}{100} \cdot \frac{3}{\text{min}}$

+ 3L/min add
− 3L/min remove

$\Delta$ 0 change

constant amount of 100L of solution

Pipe A: 10g/min

Pipe B: 10g/min

Equation in words:

Rate of change of salt = Salt Pipe A + Salt Pipe B − Salt leaving